Mask2Map: Vectorized HD Map Construction Using Bird's Eye View Segmentation Masks

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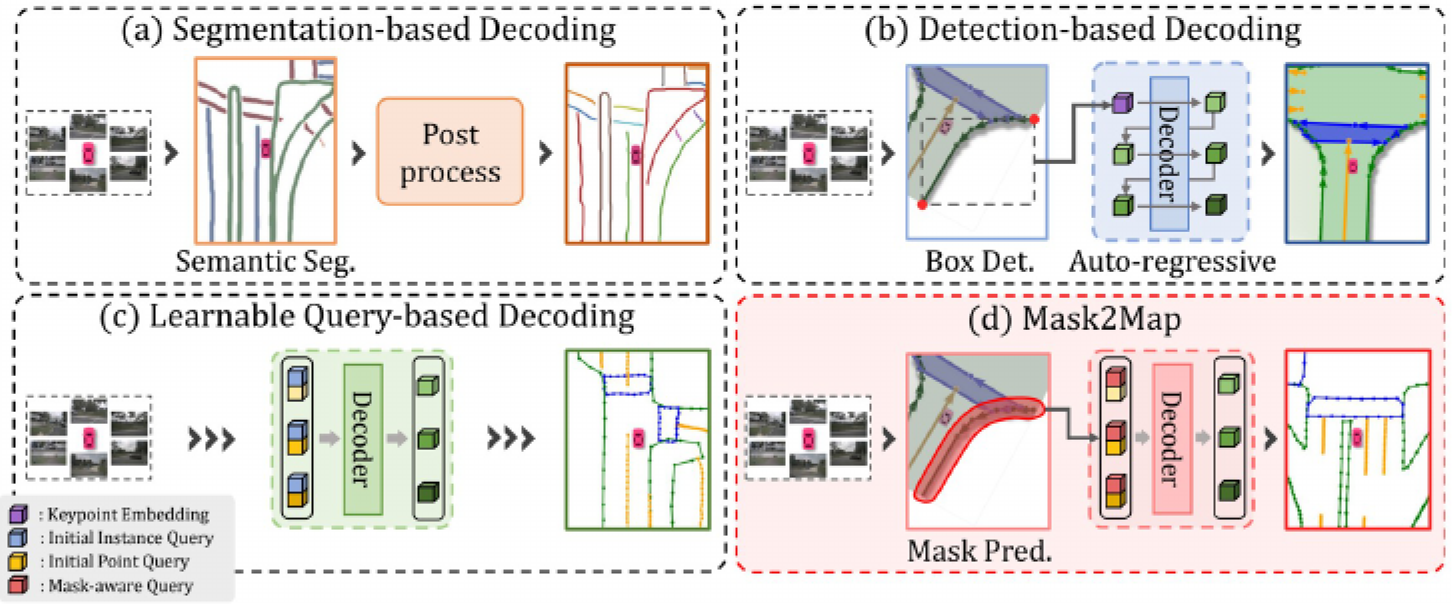
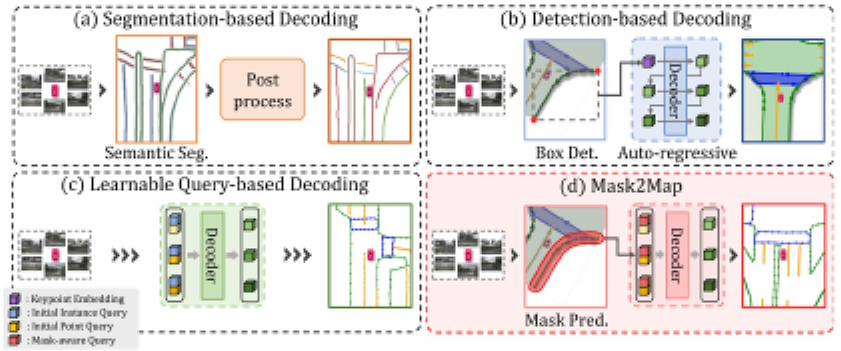
01 Why Online Vectorized HD Maps?



- Autonomy needs timely, structured HD maps with precise topology
- Offline SLAM maps are costly to build and quickly become stale
- BEV raster segmentation lacks vector structure and instance topology

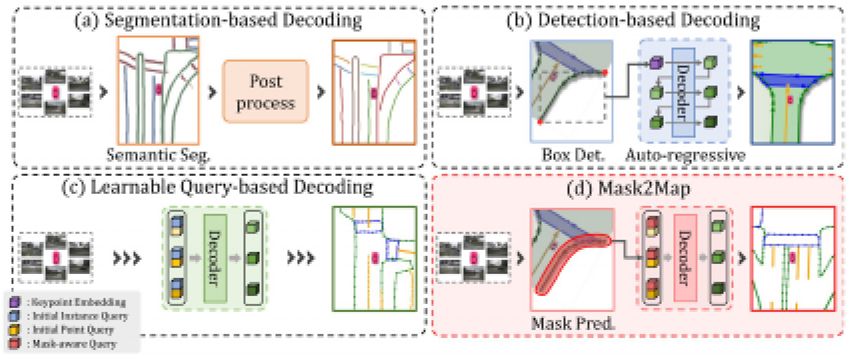
01 Limitations Of Existing Paradigms



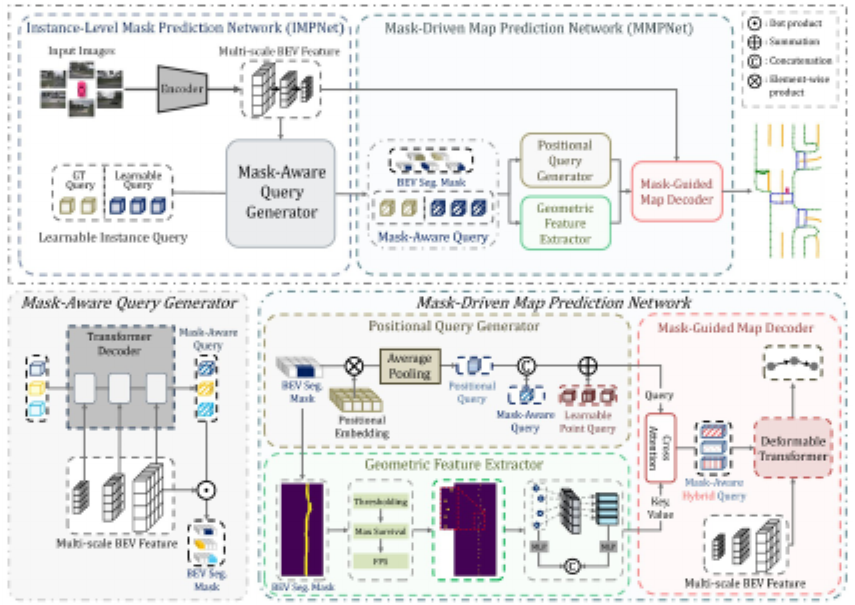
- Segmentation-plus-postprocessing is slow and brittle
- Detection keypoints fail on complex shapes and long contours
- Scene-agnostic learnable queries miss joint semantics+geometry
- Need mask-grounded queries enriched with positional and geometric cues

02 Core Contributions

- Mask-Aware Queries and BEV masks inject global semantic context
- PQG and GFE encode instance positions and point-wise geometry
- Inter-network Denoising aligns IMPNet and MMPNet for consistency
- Improves convergence and vectorized map quality



02 System Overview



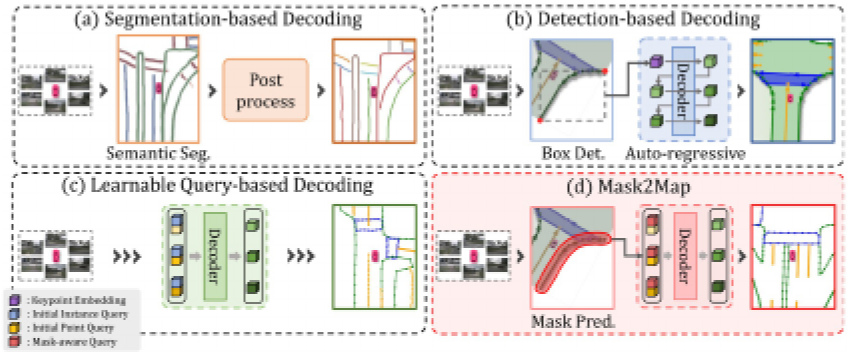
- Two networks: IMPNet produces masks and mask-aware queries
- MMPNet refines via PQG (positions) and GFE (geometry)
- Mask-Guided Decoder outputs classes and ordered BEV points

03 BEV Segmentation Background

- LSS, CVT, BEVFormer, BEVSegFormer learn dense BEV semantics
- Raster outputs lack vector topology required for planning
- Further vectorization/post-processing typically needed

03 Vectorized HD Map Construction

- Prior: HDMapNet, VectorMapNet, MapTR/MapTRv2, MapVR, PivotNet
- Different pipelines: postprocess segmentation, detect+autoregress, query decoding
- Mask2Map grounds queries in instance masks coupling semantics+geometry



03 Denoising Training Strategies

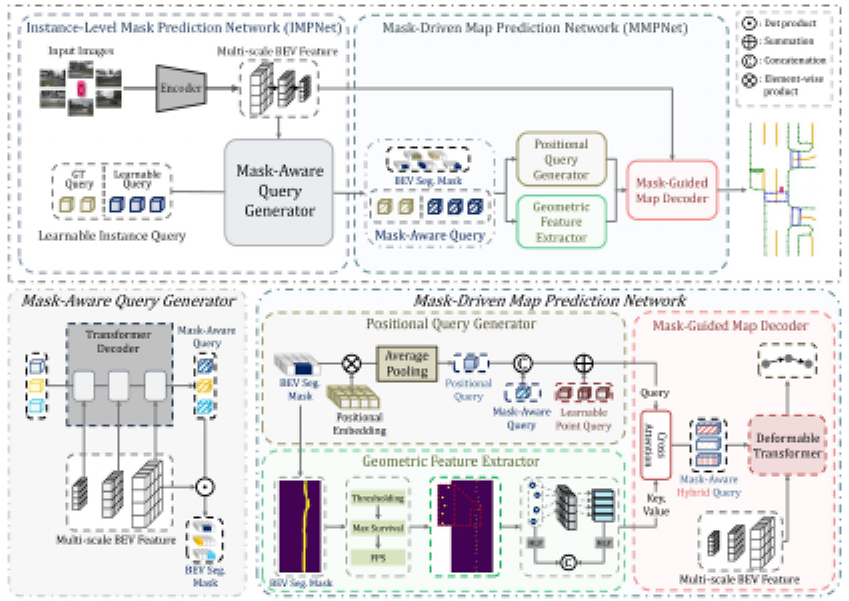
- DN-DETR uses noisy GT boxes to stabilize matching
- Mask DINO and MP-Former extend to masks and multi-task
- Mask2Map adapts denoising to align IMPNet–MMPNet correspondences

04 BEV Encoder For Multi-Scale Features

- Inputs: multi-view cameras via LSS; optional LiDAR via voxels
- Fusion by concatenation and convolution
- Deformable Transformer aggregates multi-scale BEV features
- Produces feature pyramid for mask-aware decoding

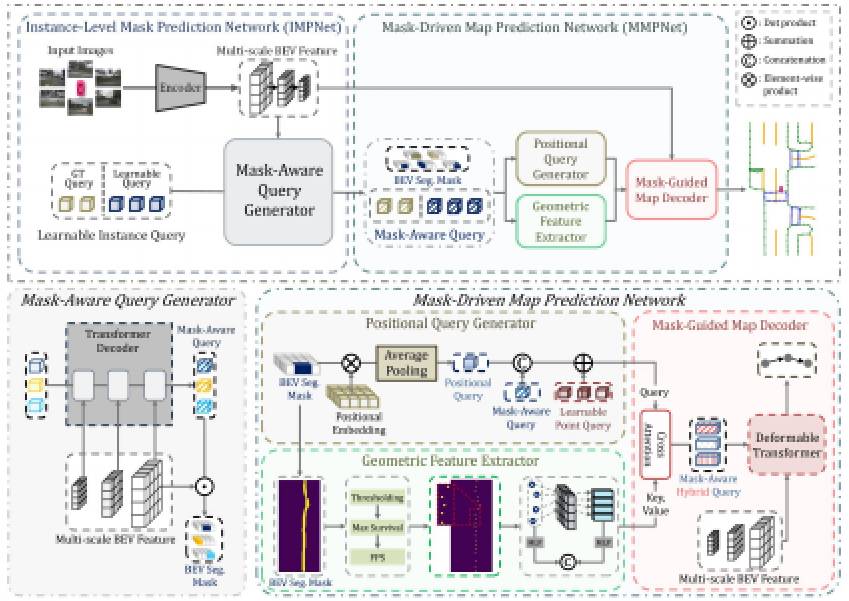
04 Mask-Aware Query Generator

- Mask Transformer iteratively refines learnable queries
- Each layer predicts BEV masks guiding next decoding steps
- Outputs final Mask-Aware Queries and refined instance masks



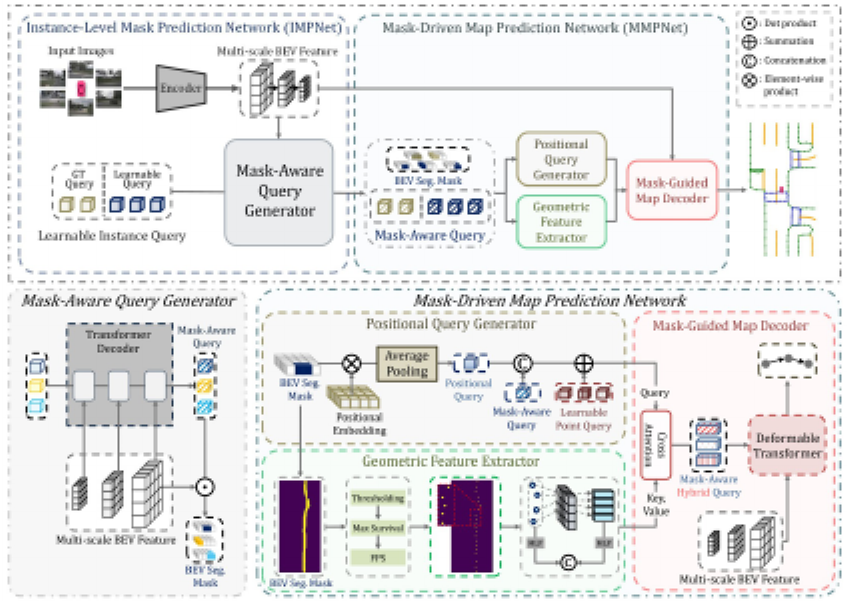
05 Positional Query Generator (PQG)

- Sparsifies masks and injects 2D sinusoidal positions
- Averages non-zero pixels to form instance positional queries
- Concatenates with Mask-Aware Queries and expands to Np points



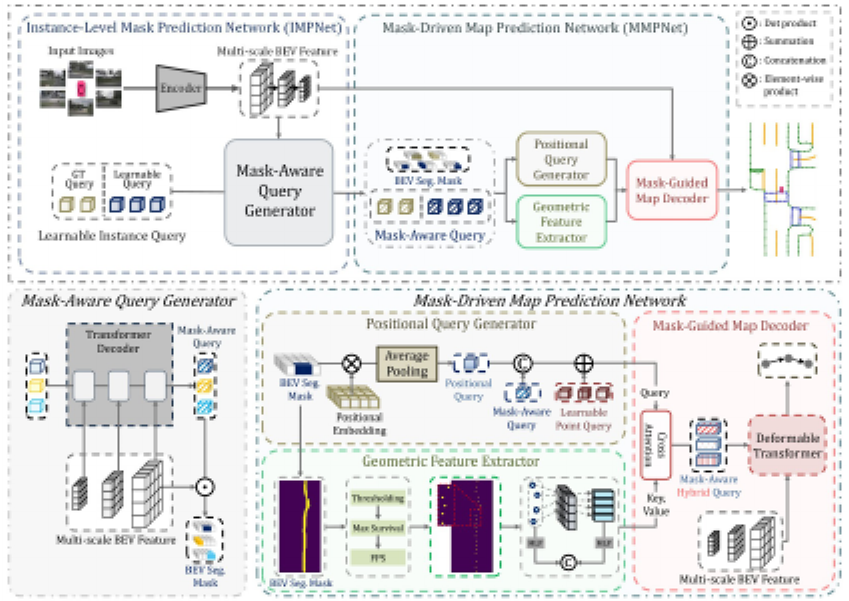
05 Geometric Feature Extractor (GFE)

- Thresholds masks and applies Max Survival over GxG windows
- Farthest point sampling selects Ns key pixels
- Pools BEV features; encodes (x,y) via MLP for geometry



05 Mask-Guided Map Decoder

- Cross-attention fuses PQG queries with GFE keys/values
- Deformable Transformer decodes over multi-scale BEV features
- Heads predict classes and normalized ordered BEV coordinates



06 Inter-Network Denoising Training For Consistency

Problem: Inconsistent Matching Across Networks

Bipartite matching mismatches across networks/epochs cause instability

Target: enforce consistent correspondences to stabilize training

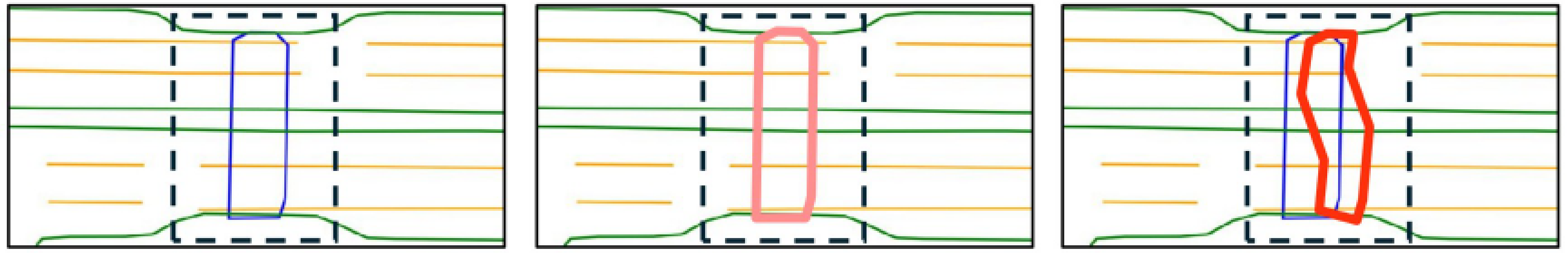
Step 1: Noisy GT Query Injection

Create GT queries from class embeddings; flip class with probability λ

Merge with learnable queries; supervise without matching

Enforces stable GT correspondences across IMPNet and MMPNet

06 Step 2: Perturbed GT Segmentation Masks



(a) (b) (c)

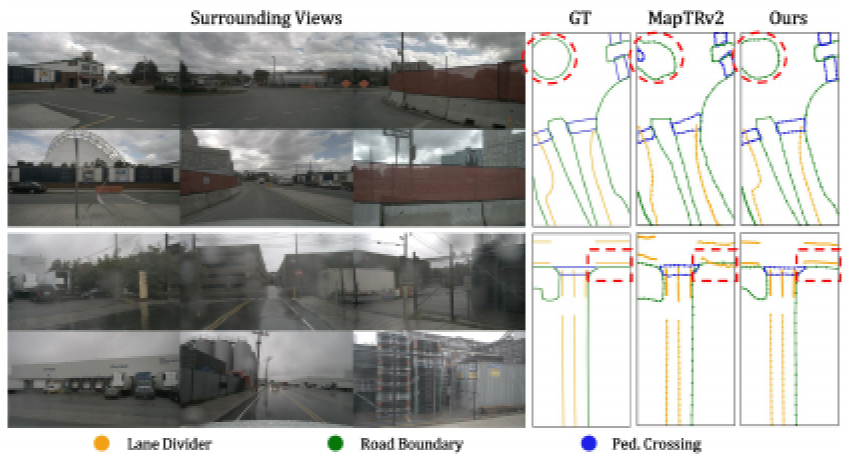
- Rasterize perturbed GT points to noisy masks (Map Noise)
- Use noisy masks to replace predicted masks for noisy GT queries
- Teaches denoising and robust mask–query associations

06 Losses And Supervision

- IMPNet: BEV segmentation loss; auxiliary depth/2D semantics
- MMPNet: map regression/classification/direction losses
- Denoising losses without matching for noisy GT queries
- Hungarian matching retained for learnable queries

07 Datasets And Metrics

- Datasets: nuScenes and Argoverse2; three map classes
- Metrics: Chamfer mAP at 0.5/1.0/1.5 m; rasterized IoU mAP
- Query Utilization measures cross-network matching consistency



07 Implementation Details

- Backbones: ResNet-50, LSS; LiDAR: SECOND
- Configs: 6 BEV encoder, 3 mask transformer, 6 decoder layers
- Hyperparams: 50 instance, 20 point queries; G=4; Ns=20; λ=0.2
- Optimization: AdamW, cosine annealing; 4×RTX3090 GPUs

08 State-Of-The-Art On nuScenes

Method	Modality	Backbone	Epoch	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP	FPS
MapTR [20]	C	R50	24	46.3	51.5	53.1	50.3	15.1
MapVR [40]	C	R50	24	47.7	54.4	51.4	51.2	15.1
PivotNet [9]	C	R50	24	56.2	56.5	60.1	57.6	-
BeMapNet [30]	C	R50	30	57.7	62.3	59.4	59.8	-
MapTRv2 [21]	C	R50	24	59.8	62.4	62.4	61.5	14.1
Ours	C	R50	24	**70.6**	**71.3**	**72.9**	**71.6**	10.1
VectorMapNet [24]	C	R50	110	36.1	47.3	39.3	40.9	2.2
MapTR [20]	C	R50	110	56.2	59.8	60.1	58.7	15.1
MapVR [40]	C	R50	110	55.0	61.8	59.4	58.8	15.1
MapTRv2 [21]	C	R50	110	68.1	68.3	69.7	68.7	14.1
Ours	C	R50	110	**73.6**	**73.1**	**77.3**	**74.6**	10.1
VectorMapNet [24]	C+L	R50 & PP	110	37.6	50.5	47.5	45.2	-
MapTR [20]	C+L	R50 & Sec	24	55.9	62.3	69.3	62.5	6.0
MapVR [40]	C+L	R50 & Sec	24	60.4	62.7	67.2	63.5	6.0
MapTRv2 [21]	C+L	R50 & Sec	24	65.6	66.5	74.8	69.0	5.8
Ours	C+L	R50 & Sec	24	**76.5**	**76.6**	**82.1**	**78.4**	4.1

- Camera-only: 71.6% mAP (24e) and 74.6% (110e), > MapTRv2
- Fusion (C+L): 78.4% mAP, +9.4 over MapTRv2
- Large rasterized mAP gains indicate strong instance masks

08 State-Of-The-Art On Argoverse2

Method	Epoch	$AP^{\dagger}_{ped}$	$AP^{\dagger}_{divider}$	$AP^{\dagger}_{boundary}$	$mAP^{\dagger}$
MapTR [20]	24	32.4	23.5	17.1	24.3
MapVR [40]	24	37.5	33.1	23.0	31.2
MapTRv2* [21]	24	49.9	34.7	25.7	36.7
Ours	24	**62.9**	**52.3**	**48.9**	**54.7**

- 71.5% Chamfer mAP with cameras and R50, +4.1 over MapTRv2
- 54.7% rasterized mAP; consistent gains across classes

09 Component Contributions

	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP
Baseline	30.1	41.5	46.6	39.4
+IMPNet	42.7	45.3	48.1	45.3
+MMPNet	45.7	47.0	54.4	49.1
+Denoising	**52.9**	**55.5**	**58.5**	**55.6**

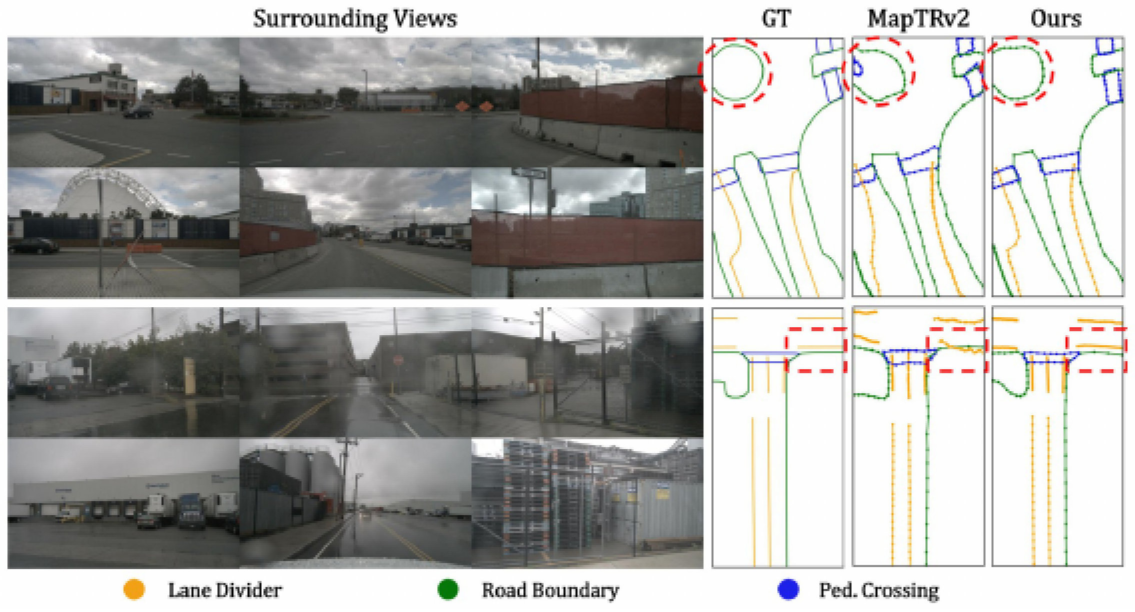
- IMPNet: +5.9 mAP; MMPNet (PQG+GFE): +3.8 mAP
- Inter-network Denoising: +6.5 mAP overall
- PQG and GFE complementary; jointly +4.8 mAP

09 Matching Consistency And Map Noise

PQG	GFE	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP
		47.6	51.1	53.8	50.8
	✓	49.7	54.2	57.8	53.9
✓		49.9	**55.6**	57.2	54.2
✓	✓	**52.9**	55.5	**58.5**	**55.6**

- Query Utilization rises from 24.7% to 74.7% with denoising
- Map Noise on GT masks adds +0.8 mAP robustness
- Improved alignment boosts overall performance

09 Qualitative Comparisons



- Sharper lane dividers and road boundaries than MapTRv2
- Better-shaped pedestrian crossings and completeness
- Highlights benefits of mask-guided semantics and geometry

10 Conclusion, Limitations, And Future Directions

Conclusion

Mask2Map fuses mask-aware semantics with positional and geometric decoding

Inter-network denoising ensures consistent training

New SOTA on nuScenes and Argoverse2 validates the design

Limitations And Future Work

Single-frame; trades some FPS for accuracy

Plan: temporal fusion over BEV/queries for occlusion robustness

Compression/architectural optimizations for real-time without accuracy loss

THANKS!